MILITARY SPECIFICATION

RADIO SET: AN/PRC-68A

This specification is approved for use by the U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for one type of lightweight, reliable, handheld pocket size radio set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	-Treatment and Painting of Materiel
MIL-E-16400	-Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	-Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts), Packaging of

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D. C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-167	-Mechanical Vibrations of Shipboard Equipment
MIL-STD-188	-Military Communication System Technical Standards
MIL-STD-454	-Standard General Requirements for Electronic Equipment
MIL-STD-461	-Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-470	-Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	-Maintainability Demonstration
MIL-STD-781	-Reliability Tests Exponential Distribution
MIL-STD-785	-Reliability Program For Systems and Equipment Development and Production
MIL-STD-810	-Environmental Test Methods and Engineering Guidelines

HANDBOOKS

MIL-HDBK-217	-Reliability Prediction of Electronic Equipment
MIL-HDBK-472	-Maintainability Prediction

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

MARINE CORPS

SMB523304	Lanyard
588005	Transducer, Permanent Magnet - AN/PRC-68
808234-1	Tool, Alignment
914161-803	Antenna, Short (Low Profile)
932831	Interface, Mechanical
977920	Procedure, Electrical Test
978122	Procedure, Reliability - Production Burn-in Test
978124	Procedure, Reliability - Production Sampling Test
978375	Procedure, Mechnical Inspection
978376	Procedure, Environmental Test
978382	Procedure, Acceptance Test

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PUBLICATIONS

NACSEM 5100 Compromising Emanations Laboratory Test Standard Electromagnetics

NACSEM 5112 Non-Stop Evaluation Techniques

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD-adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI STD S3.7-1973 -Coupler Calibration of Earphones, Method For

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 -Packaging, Commercial

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadephia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), prior to submission of any completed radio sets for acceptance, the contractor shall furnish ten first articles for determination of compliance to this document. Examination and testing shall be as specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply a product which conforms to this document. Any changes or deviations of production units from the first articles shall be subject to the approval of the contracting officer.

3.2 Parts, materials, and processes. Unless otherwise specified herein, all parts, materials, and processes used in the

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construction of this radio set shall conform to MIL-E-16400. Microcircuit devices shall be selected in accordance with requirement 64 of MIL-STD-454. The selection of class, grade, or type shall be such that, when mounted in the radio set, the part or material will perform its intended function under the environmental conditions specified in 3.6.

3.3 Design and construction. The radio set shall be in accordance with the provisions of MIL-E-16400, except as otherwise specified herein. The radio set shall comprise a complete operating radio communications terminal capable of very high frequency (VHF)/frequency modulation (FM) voice communications in the 30 to 80-megahertz (MHz) range. When interfaced with the proper equipment, the radio set shall also provide secure voice communications. The receiver-transmitter (RT) shall be of plug-in modular construction. Each radio set shall consist of the following components:

Item	Quantity
Receiver-Transmitter, Radio RT-1113A/PRC-68	1
Tool, Alignment (Drawing No. 808234-1)	1
Lanyard (Drawing No. SMB523304)	1
Antenna, Short (Low Profile) (Drawing No. 914161-803)	1

Batteries, BA-1588/U (Mercuric Oxide), BA-5588/U (Lithium), and BB-588/U (Nickle-Cadmium) are used with the radio set. Batteries are not required to be furnished with first article or production radio sets.

3.3.1 RT. The RT shall consist of a one-piece unit consisting of control head with electronic module section attached directly beneath, and a separate detachable battery compartment which fastens to the electronic module section by means of two quick-release Nielsen-type fasteners externally mounted. The electronic module section shall consist of a synthesizer/audio frequency (AF) module and a radio frequency (RF)/intermediate frequency (IF) module. All external controls except the push-to-talk (PTT) switch shall be located on the top surface of the RT unit. The PTT switch shall be located for ease of operation by both left- and right-handed operators. An internal microphone/speaker shall be located on the larger side surface near the top of the RT unit. The antenna attachment point shall be located on the left top of the control head of the RT unit. An eyelet shall be provided on one corner of the top edge of the equipment for use with a lanyard. An audio connector, compatible with handset H-250, shall be located on the right top of the control head and shall be provided with a removable dust cover. Pin E of this connector shall be used as an external +15 volts direct current (V_{dc}) power interface for vehicular installations. The case shall be designed for extremely harsh field usage. Since the operators of this equipment can be expected to be taking such actions as falling to the

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ground, rolling over and crawling, the case shall be designed so that there are no sharp edges or protrusions which could cause injury or discomfort or in any way hinder freedom of movement or individual use of weapons. The RT shall include provisions for the storage of a tuning tool on the same surface as the battery and secure voice module (SVM) connectors.

3.3.1.1 Controls and indicators. The following operating controls shall be provided on the external surface of the RT control head.

(a) Push-to-talk Switch.
(b) Volume Control.
(c) POWER OFF/POWER ON/SQUELCH DISABLE SWITCH (The switch shall not remain in the squelch-off position unless manually held in squelch-off position).
(d) PHONE/MIC CONNECTOR, 5 PIN OR 6 PIN COMPATIBLE with U-229/U.
(e) Microphone/Speaker.
(f) Channel Selector Switch (10 Position).

3.3.1.2 Communication range. A reliable communications range between two radio sets is 300 meters (328 yards) with standard low profile (short) antenna and 1.6 kilometers (km) (1 mile) or more with the blade portion (flexible base removed) of the AT-892/PRC-25 Antenna over a wide range of terrain types and with operators in supine, kneeling, and standing positions.

3.3.1.3 Hours of operation. The RT with an appropriate battery, such as the BA-1588/U, shall be capable of 36 continuous hours of operation under an operating duty cycle of 1 minute transmit, 1 minute receive, and 8 minutes standby. The standby mode is defined as the condition where the RT power control is in the "on" position (squelch on) and no transmissions or receptions are occurring. This duty cycle shall be defined as the standard duty cycle for the purpose of quality assurance testing. Battery life shall be the only limitation barring continuous operation under standard duty cycle conditions. The RT shall incorporate a low-battery detection and warning function which informs the operator that the battery is nearing end-of-life (EOL). The detection threshold shall be an internal adjustment. The low battery detection function shall be enabled only when the RT is in the standby (squelch on) mode. The radio set shall incorporate a power saving circuit which will deactivate nonessential portions of the RT for a nominal 150-millisecond (ms) period and then reactivate for a nominal 50-ms listening period. The total cycle shall be 220 ms maximum but the listening period shall not be less than 45 ms. Operation of the RT shall be returned to normal upon receipt of an incoming signal or activation of the PTT switch.

3.3.1.4 Modulation. The RT shall be designed for the transmission and reception of voice-modulated RF transmissions with a

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nominal frequency deviation of 8 kilohertz (kHz) and a maximum rated system deviation of 15 kHz. With the appropriate ancillary equipment attached, the radio set shall receive and transmit wideband data signals with a deviation of 11 kHz peak-to-peak (±5 kHz nominal).

3.3.1.5 Frequency. The RT shall incorporate a digital frequency synthesizer capable of providing operation throughout the frequency range of 30-80 MHz in 25-kHz steps. A nonvolatile preset-channels memory shall be incorporated which provides the operator the selection of ten independent channel frequencies located anywhere within one of four internally selected operating bands. The operating bands shall be 30-40 MHz, 40-54 MHz, 50-64 MHz, and 60-80 MHz. Frequency selection and preset memory programming shall be done internally with manual switches. No special RF alignment shall be required when programming operating frequencies other than setting internal band switches for the antenna matching network and the transmitter harmonic filters. The RF preselector amplifier shall be voltage tuned with a control voltage derived from the frequency synthesizer. No more than 25 degraded channels shall be allowed and shall be subject to the following conditions. No more than 25 channels shall require an RF input greater than 0.71 closed-circuit microvolts to establish 10 decibel (dB) SINAD at the receiver narrowband output. Of these 25 channels, no more than 10 channels shall require an RF input greater than 3.16 closed-circuit microvolts to establish 10-dB SINAD at the receiver narrow-band output.

3.3.1.5.1 Stability. The overall frequency stability of the RT shall not exceed 50 parts per million (ppm) (±0.005 percent) in accordance with MIL-STD-188 for low performance equipment over the environmental extremes specified for this radio set.

3.3.1.5.2 Aging. The RT frequency accuracy shall not be degraded more than 10 ppm due to aging of the frequency-determining components (crystals) over a 10-year period. Provision shall be made for manual adjustment of the reference oscillator frequency to compensate for frequency drift due to aging of components.

3.3.1.6 High RF protection. The RT shall incorporate protective circuitry to prevent damage in the event the radio set is positioned in close proximity (0.91 meter (m) (3 feet)) to the antenna of a transmitter radiating a minimum of 35 watts (W) RF anywhere within the frequency range of 30 to 80 MHz.

3.3.1.7 Vehicular power. The RT shall incorporate a direct current (DC) power connection on Pin E of the audio connector which allows the RT to be powered from a vehicular power source of +15 Vdc ±1 Vdc when the RT power switch is in the "off" position. This function shall be disabled when the RT power switch is in the "on" or "squelch disable" positions.

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3.3.2 Receiver. The design of the receiver shall be such that its performance requirements shall be met throughout the specified RT frequency range. A super-heterodyne type design shall be employed utilizing a dual IF system.

3.3.2.1 Sensitivity. The sensitivity of the receiver shall be such that a narrow band signal 0.5 microvolt or less (closed circuit) modulated with 1000 hertz (Hz) at 8 kHz deviation will produce a 10-dB SINAD (signal plus noise plus distortion to noise plus distortion) ratio measured at a receiver narrow-band audio output and shall produce at least one-half of specified audio output. The sensitivity of the receiver in the wideband mode shall be such that a 1.8 microvolt (closed circuit) signal modulated with 8000 Hz at 5 kHz (nominal) deviation shall produce a 10-dB SINAD ratio measured at a wideband audio output and shall produce at least one-half of the specified output level. Wideband tests are not required during burn-in and reliability testing. The sensitivity may be 3 dB below the 10 dB limit when subjected to testing at the temperature extremes.

3.3.2.2 Audio squelch sensitivity. A squelch circuit shall be incorporated in the receiver to eliminate noise during standby periods. This circuit shall feature internally adjustable threshold sensitivity. Maximum threshold sensitivity shall be 0.5 microvolt using a test signal of 8 kHz deviation at 1000 Hz. It shall not be possible to lock out the audio on a 5 microvolt (closed circuit) on-channel signal with the squelch control adjusted to any position.

3.3.2.3 Adjacent channel selectivity and desensitization. The output of the radio receiver shall be terminated in a standard output load. Two signal generators shall be equally coupled to the receiver antenna input terminals through a suitable matching network. Signal generator No. 1 shall be adjusted on channel for 12-dB SINAD frequency modulated 9 kHz at 1000 Hz. Signal generator No. 2, frequency modulated 8 kHz at 400 Hz, shall be tuned first to the high and then to the low adjacent channel. Signal selectivity shall be specified as the ratio, expressed in dB, of the amplitude of signal No. 2 to signal No. 1. If the ratio for the high side adjacent channel is different from the ratio for the low side adjacent channel, the smaller ratio, expressed in dB, shall be used in specifying selectivity. For specified channel of 50 kHz, the adjacent channel rejection shall be at least 55 dB.

3.3.2.4 Response to spurious signals. Spurious response attenuation and IF rejection shall be at least 60 dB relative to the measured receive sensitivity (50 dB in the range from 72 MHz to 148 MHz) except for the following.

3.3.2.4.1 Image rejection. The receiver shall attenuate signals on the image frequencies at least 40 dB relative to receiver sensitivity.

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3.3.2.4.2 Intermodulation susceptibility. The receiver shall attenuate intermodulation products at least 50 dB relative to receiver sensitivity.

3.3.2.5 Audio power output. The receiver shall be capable of delivering a minimum of 20 milliwatts (mW) at 1000 Hz with 8 kHz deviation at less than 10 percent distortion to an external load equivalent to the speaker's internal impedance. The wideband audio output shall be a minimum of 100 millivolts (mV) root mean square (rms) at 1000 Hz with 5 kHz deviation at less than 10 percent distortion to an external load greater than or equal to 20,000 ohms. The RF input signal level shall be 1.0 mV.

3.3.2.6 Audio frequency response. Narrowband audio frequency response shall not vary more than +1 or -4 dB from 300 to 3000 Hz relative to the response at 1000 Hz. Wideband audio frequency response shall not vary more than +1 or -3 dB at 10 Hz and 12 kHz, ±1 dB at 100 Hz, and +1 or -2 dB at 10 kHz relative to the response at 1000 Hz.

3.3.2.7 Hum and noise ratio. The hum and noise level in the unsquelched condition shall be at least 35 dB below rated audio power output. In the squelched condition, it shall be at least 50 dB below rated audio power output. This ratio is a measure of residual audio output to rated audio output and shall be taken at a signal level of 1 mV with and without modulation of 8 kHz deviation at 1000 Hz.

3.3.2.8 Limiting. A limiter circuit shall be provided so that with an on-channel signal applied of 8 kHz deviation at 1000 Hz and varying in amplitude from 0.8 microvolts to 50 mV (closed circuit), the receiver audio output shall not vary more than 3 dB.

3.3.2.9 Receiver response time. The receiver attack time shall be defined as the time required for the wideband audio output to reach 90 percent of steady state level as measured from time (t_o) at the start of the battery saver (chopped B+) ON cycle after a test signal of 0.5 microvolt, modulated at 8 kHz deviation with 1000 Hz, has been applied to the receiver input. The receiver attack time shall not exceed 15 ms. The receiver shall be in the squelched mode prior to application of the test signal. The normal mode squelched receiver attack time (squelch release time) shall be defined as the time required for the narrowband audio output to reach 90 percent of steady state level as measured from time (t_o) when a test signal of 0.5 microvolt, modulated at 8 kHz deviation with 1000 Hz, is applied to the receiver input. The squelch release time shall not exceed 200 ms.

3.3.3 Transmitter. The design of the transmitter shall be such that its performance requirements shall be met throughout the specified RT frequency range.

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3.3.3.1 RF power output. The RF power delivered to a 50-ohm load with the antenna coupler bypassed shall not be less than 1 W when voltage input is set for 14.5 Vdc. When subjected to testing at temperature extremes, the power may be 1 dB below 1 W. Additionally, the transmitter, when utilizing an external power source, shall be capable of a power output not less than 1 W for a period of one-half hour without damage or performance degradation. The current drain during continuous transmission shall not increase over initial transmitter current drain by more than 10 percent. The transmitter shall be protected from damage due to antenna mismatch caused by an infinite voltage standing wave ratio (VSWR) presented at the antenna connection.

3.3.3.2 Spurious and harmonic radiation. All spurious and harmonic radiations shall be attenuated at least 50 dB below the RF carrier level, with the exception of the second harmonic which shall be attenuated at least 40 dB in accordance with MIL-STD-188.

3.3.3.3 Frequency stability. The carrier frequency stability shall meet the requirements specified in 3.3.1.5.1.

3.3.3.4 Audio frequency harmonic distortion. Audio frequency harmonic distortion shall not exceed 10 percent at 8 kHz deviation with a modulating frequency of 1000 Hz in the narrowband mode. In the wideband mode, audio frequency harmonic modulation distortion shall not exceed 10 percent at 5 kHz deviation with a modulation frequency of 8000 Hz. Wideband tests are not required during burn-in and reliability testing.

3.3.3.5 Audio frequency response. Narrowband audio frequency response shall not vary more than +1 or -4 dB from 300 to 3000 Hz relative to the response at 1000 Hz. Wideband audio frequency response shall not vary more than +1 or -3 dB at 10 Hz and 12 kHz, ±1 dB at 100 Hz, and +1 or -2 dB at 10 kHz relative to the response at 1000 Hz.

3.3.3.6 Modulation limiting (narrowband). A limiter or clipper circuit shall be provided which will permit 8 kHz deviation under normal speech conditions and will provide limiting so that the transmitter modulation shall not exceed 15 kHz deviation at an audio frequency input level 20 dB greater than the level for 8 kHz when tested within the specified audio range of 300 to 3000 Hz.

3.3.3.7 Low pass audio filter (narrowband). The transmitter shall incorporate a low pass audio filter or other device which shall be installed between the modulation limiter and the modulated stage. At audio frequencies between 3 kHz and 15 kHz, the low pass filter shall have an attenuation greater than the attenuation at 1000 Hz by at least 40 $\log_{10}$ (F/3) dB where F is the audio frequency in kHz. At audio frequencies above 15 kHz, the attenuation shall be at least 28 dB greater than the attenuation at 1000 Hz.

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3.3.3.8 FM hum and noise level. The ratio of residual FM (with microphone input disabled) to a modulation of 8 kHz deviation at 1000 Hz shall be 35 dB or greater as measured in a base bandwidth of 300 to 3000 Hz.

3.3.3.9 Amplitude modulation (AM) hum and noise level. The ratio of the carrier peak alternating current (AC) voltage to the DC voltage detected from the carrier shall not exceed -34 dB.

3.3.3.10 Tone modulation. The radio set shall transmit a 150 Hz (±2 Hz) tone signal, 2.5 to 3.5 kHz deviation, to ensure compatibility with radios designed to operate on tone operated squelch. Tone shall be capable of being disabled when the SVM is attached.

3.3.3.11 System end of message (EOM). The RT shall be capable of operation during the simultaneous application of receive B+ and PTT B+ through the wideband connector to allow transmission of an EOM after release of the PTT.

3.3.3.12 Transmit response time. The transmit attack time shall be defined as the time required for the transmitter output to reach 90 percent of steady state level as measured from time (t_o) when the transmitter is keyed. The transmit attack time shall not exceed 5 ms. The transmit decay time shall be defined as the time required for the transmitter output to decay to 10 percent of steady state level as measured from time (t_o) when the transmitter is unkeyed. The transmit decay time shall not exceed 5 ms.

3.3.4 Antenna. The radio set shall incorporate as standard, a short flexible, tubular antenna which shall be 12.7 centimeters (cm) to 19.7 cm (5 to 7.75 inches) in length. The overall length of the antenna and matching connector shall not exceed 24.8 cm (9.75 inches) above the transceiver panel. The antenna shall be covered with a nongloss insulating material whose color shall be nonreflective black. The standard antenna shall be easily removable from the front panel connector, which shall also accept the blade portion (flexible base removed) of Antenna AT-892/PRC-25. The radio set shall be capable of matching either of these antennas alternately after initial channel frequency presetting and band selection. Only the standard (short) antenna shall be delivered with each radio set. The antenna coupler shall provide a 50-ohm (straight thru) switch position for compatibility with standard RF test equipment and broadband 50-ohm vehicular antenna systems.

3.3.5 Battery power supply. The radio set shall be designed to operate primarily with, and provide the required hours of operation using, a BA-1588/U Battery; 16.0 voltage operating capability (V_{oc}) to 12.0 voltage end of life (V_{eol}). Radio set

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performance characteristics applicable to battery selections shall be as follows:

(a) Duty Cycle	IDCmax @ 14.5 V_{dc}
10% Receive	45 milliampere (mA) continuous
10% Transmit	330 mA continuous
80% Standby (pulsed)	34 mA for 54 ms on 26 mA for 161 ms off

(b) Open Circuit Voltage, V_{oc}
16.0 V_{dc} maximum

(c) End of Life Voltage, V_{eol}
12.0 V_{dc} minimum
in transmit mode

(d) The low battery warning tone shall be enabled when battery voltage is at end of life (V_{eol}).

3.3.6 Audio transducers.

3.3.6.1 Audio accessory. The internal speaker/microphone shall be muted when an external Handset H-189 or H-250 is attached.

3.3.6.2 Audio transducer, internal.

3.3.6.2.1 Response, transducer as microphone. The design of the internal transducer shall be such that performance as a microphone in the radio set shall be as follows:

(a) Sound pressure reference levels used shall be expressed as 2×10^{-5} Newton per square meter or 0.0002 microbar. (Previous equivalent 0.000204 dynes per cm^2 or 10^{-12} watt/$meter^2$ energy.)

(b) Microphone Sensitivity. When illuminated by an isotropic sound pressure field at a level of 85 dB of pink noise, the radio set shall produce at least 50 percent of the nominal modulation deviation specified in 3.3.1.4.

(c) The distance from the generated sound source shall be 15.2 cm (6 inches) measured normal to the surface of the radio set upon which the transducer is mounted, and the plano-centric point of the sound source. The surrounding area shall be essentially anechoic.

3.3.6.2.2 Response, transducer as a loudspeaker. The design of the transducer shall be such that its performance as a loudspeaker in the radio set shall be as follows:

(a) When the acoustic transducer is used as a loudspeaker, the sound pressure level shall be measured at a point 15.2 cm (6 inches) normal to the surface of the radio set upon which the transducer is fixed.

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(b) The measuring device shall be essentially uniform in response from 300 Hz to 3000 Hz and shall be provided with a response indicator calibrated in dB.
(c) When an on-channel signal is supplied from a signal generator modulated with pink noise and 10 kHz deviation, with the radio set volume control at maximum, the output sound pressure level produced by the loudspeaker shall be not less than 68 dB.

3.3.6.2.3 Frequency response and distortion. The frequency response and distortion of the transducer as measured at the component level shall be as follows:

(a) The acoustic transducer shall be measured in a modified standard type I 60cc coupler, in accordance with ANSI Standard S3.7-1973 modified to precisely position the speaker opening over the acoustic cavity.
(b) The device under test shall be connected to signal source in series with a resistance equal to its nominal impedance.
(c) The signal generator output shall be adjusted to apply approximately 0.5 mW to the device under test.
(d) For 65-ohm impedance devices, this voltage is 0.175 volt (V).
(e) The sound pressure response of the loudspeaker shall fall within the limits of the curve shown in Figure 2 of Drawing No. 588005, over the frequency range of 300 to 3000 Hz.
(f) The voltage applied shall be maintained at 0.175 V at 1 kHz and the total acoustic harmonic distortion shall not exceed 15 percent. This figure does not include any disortion content contributed by the signal source or measuring apparatus.

3.3.7 Tuning tool. A tuning tool in accordance with Drawing No. 808234-1 shall be provided.

3.3.8 External connections. External connections shall be made to the front panel of the radio set. The wideband connector shall be considered an internal connection and shall be mounted in accordance with Drawing No. 932831.

3.3.9 Covers and seals. Covers for this radio set shall incorporate captivated hardware. Seals or sealing surfaces shall be capable of 100 open-and-close cycles without loss of leakage resistance, and with no seal maintenance other than gasket replacement as required by visual inspection.

3.3.10 Reverse polarity protection. The design shall incorporate mechanical and electrical reverse polarity protection.

3.3.11 SVM interface-mechanical. SVM interface-mechanical shall be in accordance with Drawing No. 932831 and Appendix A.

3.3.12 SVM interface-electrical. SVM interface-electrical shall be in accordance with Appendix A.

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3.3.13 Soldering. Soldering shall be in accordance with requirement 5 of MIL-STD-454.

3.3.14 Burn-in. Prior to submission to the specified first article and quality conformance inspections, each radio set shall be subjected to burn-in as specified in 4.6.6.

3.3.15 Interchangeability. Provisions for interchangeability are required and shall be in accordance with requirement 7 of MIL-STD-454.

3.4 Reliability. The radio set shall have a lower test mean-time-between failures (MTBF) (θ_1) of 1000 hours and an upper test MTBF (θ_0) of 2000 hours in accordance with Test IV of MIL-STD-781 except the temperature range shall be -40°C to +65°C.

3.4.1 Reliability program plan. A program plan shall be prepared, when specified (see 6.2), in accordance with MIL-STD-785. The plan shall include a failure data collection, analysis, and corrective action system.

3.4.2 Reliability prediction. A reliability prediction shall be performed and reported, when specified (see 6.2), in accordance with MIL-STD-785 and section 2 of MIL-HDBK-217. The prediction shall assume a +65 °C external ambient air temperature, considering temperature rises within the electronic enclosures and modules, and assuming a "Ground Mobile" environment.

3.5 Maintainability. The radio set shall be constructed to provide ease of maintenance, accessibility and replacement of all modules and parts as primary design considerations. Maintenance and repair shall be accomplished to the maximum extent practicable when the system is operational.

3.5.1 Maintainability program plan. A maintainability plan shall be prepared, when specified (see 6.2), in accordance with MIL-STD-470.

3.5.2 Maintainability prediction. A maintainability prediction shall be performed and reported, when specified (see 6.2), in accordance with Procedure III of MIL-HDBK-472 at each of the three levels of maintenance.

3.5.2.1 Organizational level. Corrective maintenance limited to replacement of battery and antenna. The test set TS-3354/PRC-68 shall be used in performing operational checks. The maximum time to repair shall be 15 minutes at the 95th percentile.

3.5.2.2 Intermediate level. Corrective maintenance limited to replacement of the battery and modules. The field test set TS-3354/PRC-68, maintenance kit MK-2137/PRC-68, and standard test

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equipment shall be used to perform the module fault isolation. The maximum time to repair shall be 15 minutes at the 95th percentile.

3.5.2.3 Depot level. Repairs to be made to the piece part level of repairable modules. A test set TS-3354/PRC-68, maintenance kit MK-2137/PRC-68, special subassembly test fixtures, standard test equipment, and normal bench tools shall be available. Maximum time to repair shall be 3 hours at the 95th percentile.

3.6 Performance. The radio set shall meet the requirements specified herein after exposure to environmental conditions as follows:

3.6.1 Nonoperating.

3.6.1.1 Temperature storage. The radio set shall be capable of normal operation after storage at any temperature within the range of -62°C to +72°C.

3.6.1.2 Immersion. The RT case, with and without the battery case, shall be watertight in accordance with MIL-STD-810.

3.6.1.3 Drop test. The RT shall be capable of satisfactory operation with no degradation in performance limits after completion of the test specified in 4.6.1.3. The RT, with battery installed, shall be subjected to a series of drops in accordance with MIL-STD-810. The RT shall withstand this test with no mechanical damage other than superficial case damage.

3.6.1.4 Salt spray. The radio set shall be capable of meeting salt spray test requirements specified in MIL-E-16400.

3.6.1.5. Humidity. The radio set shall be capable of meeting the humidity test requirements specificed in MIL-STD-810, Method 507.1, Procedure IV.

3.6.1.6 Fungus. The radio set shall be capable of meeting the fungus test requirements specified in MIL-STD-810.

3.6.2 Operating.

3.6.2.1 Temperature operating and nonoperating. The radio set shall be Range 1, as specified in MIL-E-16400 Table I except that the temperature operating range shall be -40°C to +65°C.

3.6.2.2 Supply line voltage. The radio set shall be designed to operate from a DC battery source of 16.0 V open-circuit and operate at supply voltages down to 12.0 V.

3.6.2.3 Vibration. The radio set shall be designed to operate satisfactorily during the vibration test specified in MIL-E-16400,

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except that the vibration range shall be of 5 to 55 Hz. The endurance test requirements of MIL-STD-167 for type 1 vibration shall apply throughout the range of 5 to 55 Hz. The amplitudes of vibration in the range of 5 to 55 Hz shall be noted by the contractor and included in the test report.

3.6.2.4 Altitude. The radio set shall be capable of continuous operation up to an altitude of 3,048 km (10,000 feet).

3.6.2.5 Sand and dust. The radio set shall be capable of operation during and after the dust test specified in MIL-STD-810, Method 510, Procedure I.

3.6.2.6 High RF field. The receiver, using blade portion (flexible base removed) of the Antenna AT-892/PRC-25, shall be capable of satisfactory operation after having been positioned in close proximity (0.91 m (3 feet)) to the antenna of a transmitter radiating a minimum of 35 W RF anywhere within the frequency range of 30 MHz to 80 MHz.

3.6.2.7 Position (inclination). The radio set shall be capable of continuous and reliable operation on an 1:1:1 standby, receive, transmit duty cycle with each of its six sides parallel and adjacent to a reference plane, normally the top of the test bench.

3.6.2.8 Humidity. The radio set shall be capable of meeting the humidity test requirements specified in MIL-STD-810, Method 507.1, Procedure IV.

3.6.2.9 Tempest. The equipment shall comply with the requirements of NACSEM 5100 and 5112 for compromising emanations related to the information processed by the radio.

3.7 Safety. Personnel hazard safety shall be in accordance with Requirement 1 of MIL-STD-454.

3.8 Electromagnetic interference (EMI). The generation of, and vulnerability to, EMI by the equipment shall meet the requirements of MIL-STD-461, Notice 1 (Table A-IV for Class SS-1 equipment), and this specification. The MIL-STD-461, Table A-IV, applicable requirements are CE06, CS08 as modified by 3.3.2.4, RE02 and RS03. The additional EMI requirements are spurious and harmonic radiation (3.3.3.2 and 4.6.7.13); intermodulation susceptibility (3.3.2.4.2 and 4.6.7.7.2); high RF field (3.6.2.6 and 4.6.1.11); high RF protection (3.3.1.7 and 4.6.7.3); and adjacent channel selectivity and desensitization (3.3.2.3 and 4.6.7.6). The tests applicable to the requirements of first article inspection (4.3) and to environmental inspection (4.4.3) are delineated in Table 1, Section 4.

3.9 Dimensions. The dimensions of the RT in its case, including controls, connectors, and battery compartments, but excluding

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battery compartment attachment fastener, shall be no larger than 23.72 cm (9.34 inches) in length by 9.6 cm (3.78 inches) in width by 3.86 cm (1.52 inches) in depth.

3.10 Weight. The complete RT, plus BA-1588/U battery, shall weigh no more than 1406 grams (50 ounces).

3.11 Finish. The radio set shall be finished in accordance with the requirements of MIL-T-704, type F or G as applicable. Exterior color shall be green 383.

3.12 Marking. Clear, concise and durable marking shall be provided for all controls, switches, test points and other components as necessary. All areas or portions of the radio which present a safety hazard to personnel shall be clearly and concisely marked in accordance with MIL-STD-454. The RT shall contain a nameplate, which shall contain blank spaces in which to identify the unit serial number, the contract number, and the contractor's Federal Supply Code for Manufacturers. The unit serial numbers will be specified by the contract (see 6.2).

3.13 Government-furnished property. The contractor will require one copy of the Operators Manual, TM-06827B-10/1 to be furnished by the Government for each radio set he produces, to be packed with the radio set for shipment.

3.14 Government-loaned property. The contractor will require the following Government-loaned property for the AN/PRC-68A to assist him in his testing and production efforts. The procurement contracting officer will arrange to loan this property to the contractor.

(a) Antenna, AT-892/PRC-25, quantity 2.
(b) Handset, H-250/U, quantity 2.
(c) Secure Voice Module, TSEC KYV-2 or -2A, quantity 2.
(d) Test Set TS-3354/PRC-68, quantity 1.
(e) Maintenance Kit, MK-2137/PRC-68, quantity 1.
(f) When specified (see 6.2), special tooling for production.
(g) Maintenance Manual, TM 06827B-24/2.

3.15 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2. Classifications of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
    (1) Production inspection (see 4.4.1).
    (2) Production control inspection (see 4.4.2).
    (3) Environmental tests (see 4.4.3).
    (4) Inspection of packaging (see 4.7).

4.3 First article inspection. The first articles shall be subjected to all of the examinations and tests indicated in Table I. When specified (see 6.2), the contractor shall submit a first article test plan for approval prior to the start of testing. The quantity of first articles to be tested shall be specified by the procurement contracting officer (see 6.2).

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following. Reference Drawing No. 978382 for the test procedures to be used.

4.4.1 Production inspection. Production inspection shall be made on every radio set offered for delivery. The inspection shall consist of examination and testing which will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests indicated in Table I.

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TABLE I. TESTS

Name	Reqts Para	Test Para	Quality Conformance 1st Art	Prod	Env
1. Maintainability	3.5	4.6.3	x		
2. Reliability	3.4	4.6.2	x		
3. Dimensions	3.9	4.6.4	x		
4. Hours of Operation	3.3.1.3	4.6.7.1	x		
5. Electromagnetic Interference	3.8	4.6.7.2	x		
6. High RF Protection	3.3.1.6	4.6.7.3	x		x
7. Sensitivity	3.3.2.1	4.6.7.4	x	x	
8. Audio Squelch Sensitivity	3.3.2.2	4.6.7.5	x	x	
9. Selectivity	3.3.2.3	4.6.7.6	x		x
10. Response to Spurious Signals	3.3.2.4	4.6.7.7	x		x
11. Image Rejection	3.3.2.4.1	4.6.7.7.1	x	x	
12. Intermodulation Susceptibility	3.3.2.4.2	4.6.7.7.2	x		x
13. Covers and Seals	3.3.9	4.6.7.27	x		
14. Audio Power Output	3.3.2.5	4.6.7.8	x	x	
15. Audio Frequency Response	3.3.2.6	4.6.7.9	x	x	
16. Hum and Noise Ratio	3.3.2.7	4.6.7.10	x	x	
17. Limiting	3.3.2.8	4.6.7.11	x	x	
18. Radio Frequency Power Output	3.3.3.1	4.6.7.12	x	x	
19. Spurious & Harmonic Radiation	3.3.3.2	4.6.7.13	x		x
20. Transmitter Frequency Stability	3.3.3.3	4.6.7.14	x	x	
21. Audio Frequency Harmonic Distortion	3.3.3.4	4.6.7.15	x	x	
22. Audio Frequency Response	3.3.3.5	4.6.7.16	x	x	
23. Modulation Limiting	3.3.3.6	4.6.7.17	x	x	
24. Low Pass Audio Filter	3.3.3.7	4.6.7.18	x	x	
25. Frequency Modulation Hum and Noise Level	3.3.3.8	4.6.7.19	x	x	
26. Amplitude Modulation Hum and Noise Level	3.3.3.9	4.6.7.20	x		
27. Tone Modulation	3.3.3.10	4.6.7.21	x	x	
28. Battery Power Supply	3.3.5.(a)	4.6.7.22	x	x	
29. Antenna	3.3.4	4.6.7.23	x	x	
30. Temperature Storage	3.6.1.1	4.6.1.1	x		x
31. Immersion	3.6.1.2	4.6.1.2	x		x
32. Drop Test	3.6.1.3	4.6.1.3	x		
33. Salt Spray	3.6.1.4	4.6.1.4	x		
34. Humidity	3.6.1.5	4.6.1.5	x		x

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Table I (cont'd)

Name	Reqts Para	Test Para	Quality Conformance 1st Art	Prod	Env
35. Temperature Operating	3.6.2.1	4.6.1.6	x		x
36. Supply Line Voltage	3.6.2.2	4.6.1.7	x		x
37. Vibration	3.6.2.3	4.6.1.8	x		x
38. Altitude	3.6.2.4	4.6.1.9	x		x
39. Sand and Dust	3.6.2.5	4.6.1.10	x		
40. High RF Field	3.6.2.6	4.6.1.11	x		
41. Position	3.6.2.7	4.6.1.12	x		
42. Interchangeability	3.3.15	4.6.5	x		x
43. Burn-In	3.3.14	4.6.6	x	x	
44. Mechanical Inspection	3.2	4.5	x	x	
45. Audio Transducer Internal	3.3.6.2	4.6.7.24	x		
46. SVM Interface-Mechanical	3.3.11	4.6.7.25	x	x	
47. SVM Interface-Electrical	3.3.12	4.6.7.26	x	x	
48. Inspection of Packaging	5.1	4.7		x	
49. Fungus Test	3.6.1.6	4.6.1.13	x		
50. Weight	3.10	4.6.4	x		
51. Receive Response Time	3.3.2.9	4.6.7.28	x		
52. Transmit Response Time	3.3.3.12	4.6.7.29	x		
53. Audio Accessory	3.3.6.1	4.6.7.30	x	x	
54. Tempest	3.6.2.9	4.6.1.14	x		
55. EOM	3.3.3.11	4.6.7.26	x	x	
56. Vehicular Power	3.3.1.7	4.6.7.31	x	x	

4.4.2 Production control inspection. Production control (P/C) inspection, including sampling, shall consist of Test No. 2 (Reliability) indicated in Table I and the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. The acceptable quality level (AQL) shall be 6.5 percent. A lot is defined as one month's production of radio sets. Production control inspection shall be performed on radio sets which have passed production inspection. The radio sets shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

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4.4.3 Environmental inspection. Environmental inspection shall consist of tests indicated in Table I and shall be performed on units which have been subjected to and have passed production inspection. Reference Drawing No. 978376 for the test procedures to be used.

4.4.3.1 Sampling for environmental inspection. Three radio sets shall be selected from each successive lot produced. A lot is defined as one month's production of radio sets. The first sample lot radio sets shall be selected from the first month's production. Environmental tests shall be distributed within each sample of three radio sets as follows:

EQUIPMENT	TEST NUMBERS
Radio No. 1	30, 34, 35
Radio No. 2	31, 37, 38, 42
Radio No. 3	36, 6, 42, 19, 10, 9, 12

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the administrative contracting officer the results thereof and details of the corrective action taken to correct units of product that were manufactured under the same conditions with same materials, processes and so forth. If the administrative contracting officer does not consider that the corrective action will enable the product to meet the requirements specified herein, or if the contractor cannot determine the cause of failure, the matter shall be referred to the procurement contracting officer.

4.4.3.3 Reinspection of conforming environmental sample units. Sample units which have been subjected to, and passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.5 Inspection. Inspection shall be performed to determine conformance to the requirements specified herein prior to inspection of packaging. Radio sets shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and in 3.2. Reference Drawing No. 978375. Particular attention shall be given to the following:

(a) Completeness.
(b) Nameplates, identification marking and labels.
(c) Ease of operation of all front panel controls and switches and all connectors. During first article testing, all connectors

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will be physically mated with the Government-loaned property (see 3.14) and the battery (see 3.3) to verify the mechanical interfaces.

(d) Finishes.
(e) Welded joints, if any.
(f) Solder joints.
(g) The fit of components, to include the SVM, in their respective positions.
(h) Battery compartment fasteners.
(i) Fit and condition of seals.
(j) Other visual defects.

4.6 Tests.

4.6.1 Environmental. Where reference measurements are required for a particular environmental test, the following performance tests shall be performed at an operating frequency of 43.05 MHz:

Test No. 7	Sensitivity
Test No. 14	Audio Power Output
Test No. 18	Radio Frequency Power Output
Test No. 20	Transmitter Frequency Stability
Test No. 21	Audio Frequency Harmonic Distortion

4.6.1.1 Temperature storage. The temperature storage test shall be performed as part of the temperature test specified in MIL-E-16400.

4.6.1.2 Immersion. The RT, with and without battery case, shall be subjected to the immersion test specified in MIL-STD-810, Method 512.1, Procedure I.

4.6.1.3 Drop. The radio set shall be subjected to a series of drops from a height of 106.7 cm (42 inches) to a flat surface constructed of a layer of pine not more than 2.54 cm (1 inch) thick and a concrete foundation at least 5.08 cm (2 inches) thick. The radio set shall be dropped twice on each of its sides and each corner for a total of 28 drops. The radio set shall then operate satisfactorily with no degradation in performance specified herein, which shall be verified through electrical testing of the reference measurements.

4.6.1.4 Salt spray. The salt spray test shall be conducted in accordance with the requirements of MIL-E-16400, except that a complete RT shall be subjected to the test.

4.6.1.5 Humidity. The humidity test shall be performed in accordance with the requirements of MIL-STD-810, Method 507.1, Procedure IV.

4.6.1.6 Temperature operating. The temperature operating test shall be performed as part of the temperature test specified in MIL-E-16400.

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4.6.1.7 Supply line voltage. The radio set shall be operated at the voltage extremes specified in 3.6.2.2 and the results compared with data taken at the midpoint voltage. Reference measurements shall be used for this test and the following limits shall apply:

Test No. 7	Sensitivity	No more than 6 dB degradation.
Test No. 14	Audio Power Output	No more than 3 dB degradation.
Test No. 18	Radio Frequency Power Output	No more than 3 dB degradation.
Test No. 20	Transmitter Frequency Stability	Shall meet the requirements of 3.3.3.3
Test No. 21	Audio Frequency Harmonic Distortion	Shall meet the requirements of 3.3.3.4

4.6.1.8 Vibration. The vibration test shall be performed in accordance with the requirements of MIL-E-16400, less shock, inclination, and noise.

4.6.1.9 Altitude. The radio set shall be operated at prevailing ambient conditions for 1 hour during which time reference measurements shall be taken and recorded. The barometric pressure within the test chamber shall then be lowered to 52.3 cm (20.58 inches) of mercury and the radio set operated for 4 hours. Radio set performance shall be monitored and reference measurements taken. The barometric pressure shall be returned to the ambient condition and reference measurements repeated. There shall be no degradation in performance.

4.6.1.10 Sand and dust. The sand and dust test shall be performed as specified in MIL-STD-810, Method 510, Procedure 1.

4.6.1.11 High RF field. Compliance with the requirements of 3.6.2.6 shall be demonstrated at a frequency in the upper end of the frequency band, in the normal hand-held or carried position.

4.6.1.12 Position. The radio set shall be operated on a duty cycle of 1-minute transmit, 1-minute receive, and 1-minute standby for a period of 1.5 hours with each of its six sides parallel to the horizontal surface of a bench. Test 15 minutes in each position. Reference measurements shall be taken at the end of this test to show compliance to requirements specified herein.

4.6.1.13 Fungus. The fungus test shall be performed per MIL-STD-810, Method 508, Procedure I.

4.6.1.14 Tempest. Not less than four first article RT units

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shall be tested to demonstrate compliance with the requirements of NACSEM 5112. One RT unit shall be tested to demonstrate compliance with NASCEM 5100. As directed by the contract, the contractor shall submit the separate test plans to the procuring activity for approval. Approval must be granted prior to commencement of testing.

4.6.2 Reliability.

4.6.2.1 Reliability qualification test. The contractor shall conduct a reliability qualification test on a maximum of eight first article units as selected by the Government. The test shall be conducted in accordance with Test Plan IV of MIL-STD-781, except that the temperature range shall be -40°C to 65°C.

4.6.2.2 Reliability sampling test. Not less than four, nor more than ten, RT units shall be randomly selected, by the Government inspector, from every monthly production lot for testing in accordance with Drawing No. 978124. The temperature range shall be -40°C to +65°C.

4.6.2.3 Modes of operation and test parameters. The radio set shall operate continuously in an 8:1:1 standby, receive, transmit duty cycle. During the test, reference measurement specified in 4.6.1 (narrowband mode) shall be taken daily, except that the following test frequencies shall be allocated to the radio sets on test in approximately equal proportions with one frequency for each radio.

Test Frequency (MHz)	Radio Block
30.05	1
43.05	2
53.95	3
54.05	4
67.05	5
79.95	6

The temperature cycle shall be adjusted so that approximately one-half of the performance measurements are made at high temperature and one-half at low temperature. MIL-STD-781 is applicable.

4.6.2.4 Failure. A failure is defined to occur any time the equipment fails to meet the respective test parameters associated with the test. In addition, those failures will be categorized as relevant and nonrelevant as follows:

(a) Relevant failures.
1. Intermittent failures.
2. Unverified. Failures which cannot be duplicated or are still under investigation, or for which no cause could be determined.

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3. Verified. Failures not otherwise excluded under nonrelevant failures.

(b) Nonrelevant failures.

1. Installation damage.
2. Accident or mishandling.
3. Failures of the test facility or test-peculiar instrumentation.
4. Equipment failures caused by externally applied overstress condition in excess of the approved test requirements.
5. Normal operating adjustments (nonfailures) as prescribed in the approved equipment operating instructions.

(c) MIL-STD-781 shall apply.

4.6.2.5 Reliability test procedures. No single equipment "ON" time shall be less than one-half the average operating time of all equipments undergoing reliability sample testing. The contractor may use a power supply in lieu of batteries for this test procedure; however, if a power supply is utilized, the input voltage shall be maintained at 16 V_{dc} for one-third of the equipment "on" cycle, at 14 V_{dc} for the second one-third of the equipment "on" cycle, and at 12 V_{dc} for the final one-third of the equipment "on" cycle. This cycling procedure is to be repeated continuously throughout the reliability test. Reference Drawing No. 978124 for test procedures to be used.

4.6.3 Maintainability demonstration. When specified (see 6.2), a separate maintainability demonstration plan for the first articles shall be prepared by the contractor in accordance with MIL-STD-471, to satisfy the requirement of a 15 minute maximum time to repair at Organization and Intermediate levels. The consumer's risk (B) for the demonstration is 10 percent. The maximum corrective maintenance down-time (M_{maxc}) shall be taken to represent the 95th percentile. Task selection shall be in accordance with Appendix A of MIL-STD-471. For the purpose of this demonstration, a minimum of 30 corrective maintenance tasks shall be demonstrated. Tasks shall be limited to fault isolation and replacement of modules in accordance with steps (a), (b), and (c) below. When specified (see 6.2), the demonstration may contain an optional depot maintenance portion in accordance with step (d).

(a) The fault is introduced in a module.
(b) A repair technician is given the failure indication (i.e., audio is distorted, controls inoperative, etc.).
(c) A repair technician troubleshoots and replaces modules in accordance with the approved repair procedures.
(d) A repair technician will then troubleshoot and repair the radio, defective module, or component in accordance with the approved depot level repair procedures.

4.6.3.1 Test procedures. The test/trouble-shooting/repair

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procedures used by the maintenance technician during the maintainability demonstration shall be the same procedures contained in the equipment's technical manual. The use of any other procedures must have prior approval of the procuring activity. The test equipment/fixtures used during the maintainability demonstration shall be the same equipment called out in the technical manual. The technical manual shall be provided by the contracting officer as Government-loaned property (see 3.14).

4.6.3.2 Data collection. The contractor shall be responsible for data collection during the maintainability demonstration. This shall be accomplished by proper use of an accurate time recorder. The time recorder shall be capable of recording localization, isolation, disassembly, replacement, reassembly, alignment, and checkout to the nearest ±5 seconds. The times shall be recorded on appropriate data sheets.

4.6.3.3 Demonstration team. The demonstration test team, as a minimum, shall consist of:

(a) A contractor's director to be responsible for conducting the demonstration and for inducing the faults.
(b) A contractor maintenance technician to perform the repair procedures as outlined in the technical manual.
(c) A contractor data collector.
(d) Government witness.

4.6.3.4 Fault simulation. Fault simulation shall be accomplished in the following manner:

(a) Introducing an internally shorted or opened component.
(b) Introducing a similar component with a different value or rating.
(c) Introducing misalignment by detuning.
(d) Other contractor-submitted and -approved methods.

4.6.4 Dimensions and weight. The radio set shall be inspected for compliance with 3.9 and 3.10.

4.6.5 Interchangeability. The two modules of the RT shall be tested for interchangeability by substitution of like modules between main chassis and control head. An operational checkout by establishing communication between two radio sets on at least three frequencies will confirm the interchangeability of modules.

4.6.6 Burn-in. Every deliverable radio set shall be subjected to a 48-hour burn-in with the last 24 hours being failure free. The burn-in shall be conducted in accordance with Drawing No. 978122. The duty cycle shall be 8:1:1 standby, receive, transmit. The equipment shall be exercised and monitored in the same manner as during reliability testing. The burn-in failures shall be

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recorded and analyzed but shall not be considered relevant for computation of equipment MTBF. If a radio set experiences a failure prior to the completion of the 24-hour failure free period, the radio set shall be repaired and resubmitted to burn-in until it successfully completes a 24-hour failure free period. Each radio set shall pass the environmental tests specified in 4.6.1 at the completion of burn-in.

4.6.7 Performance.

4.6.7.1 Hours of operation. Compliance with the requirements of 3.3.1.3 shall be demonstrated as specified in 3.3.5.

4.6.7.2 Electromagnetic interference. Compliance with the requirements of 3.8 shall be demonstrated.

4.6.7.3 High RF protection. The receiver input shall be connected to a signal generator or power amplifier having an output of 3 V_{rms} open-circuit. The signal generator shall be tuned to the receiver frequency. After applying the signal, repeat test number 7 for demonstration of compliance with requirements of 3.3.2.1.

4.6.7.4 Sensitivity. Compliance with the requirements of 3.3.2.1 shall be demonstrated on 25 frequencies spread across the frequency range during first article testing. For production testing, narrowband sensitivity shall be measured at 6 frequencies: 30.05 MHz, 43.05 MHz, 53.95 MHz, 54.05 MHz, 67.05 MHz, and 79.95 MHz. Wideband sensitivity testing shall be conducted at one frequency, 43.05 MHz.

4.6.7.5 Audio squelch sensitivity. Compliance with the audio squelch sensitivity requirements of 3.3.2.2 shall be performed in the same manner as for sensitivity requirements in 4.6.7.4, except that production testing shall be conducted at one frequency, 43.05 MHz.

4.6.7.6 Selectivity. Compliance with the requirements of 3.3.2.3 shall be demonstrated at 43.05 MHz.

4.6.7.7 Response to spurious. Compliance to the requirements of 3.3.2.4 shall be demonstrated at 43.05 MHz and 67.05 MHz.

4.6.7.7.1 Image rejection. Compliance to the requirements of 3.3.2.4.1 shall be demonstrated at 43.05 MHz and 67.05 MHz.

4.6.7.7.2 Intermodulation susceptibility. Compliance to the requirements of 3.3.2.4.2 shall be demonstrated at 43.05 MHz and 67.05 MHz.

4.6.7.8 Audio power output. Compliance with the requirements of 3.3.2.5 shall be demonstrated at 43.05 MHz.

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4.6.7.9 Audio frequency response. Compliance with the requirements of 3.3.2.6 shall be demonstrated at 43.05 MHz.

4.6.7.10 Hum and noise ratio. Compliance with the requirements of 3.3.2.7 shall be demonstrated at 43.05 MHz.

4.6.7.11 Limiting. Compliance with the requirements of 3.3.2.8 shall be demonstrated at 43.05 MHz.

4.6.7.12 RF power output. Compliance with the requirements of 3.3.3.1 shall be demonstrated on the same 25 frequencies used for sensitivity measurements during first article testing. For production testing, RF power output shall be measured at 6 frequencies: 30.25 MHz, 43.65 MHz, 53.75 MHz, 54.25 MHz, 67.65 MHz, and 79.75 MHz.

4.6.7.13 Spurious and harmonic radiation. Compliance with the requirements of 3.3.3.2 shall be demonstrated on first article radio sets using radiated measurements. For production control radio sets, compliance with the requirements of 3.3.3.2 shall be demonstrated using conducted measurements with limits specified in Dwg. No. 978376.

4.6.7.14 Frequency stability. Compliance with the requirements of 3.3.3.3 shall be demonstrated on the same test frequencies required in 4.6.7.12.

4.6.7.15 Audio frequency harmonic distortion. Compliance with the requirements of 3.3.3.4 shall be demonstrated at 43.650 MHz.

4.6.7.16 Audio frequency response. Compliance with the requirements of 3.3.3.5 shall be demonstrated at 43.650 MHz.

4.6.7.17 Modulation limiting. Compliance with the requirements of 3.3.3.6 shall be demonstrated at 43.650 MHz.

4.6.7.18 Low pass audio filter. Compliance with the requirements of 3.3.3.7 shall be demonstrated at 43.650 MHz.

4.6.7.19 FM hum and noise level. Compliance with the requirements of 3.3.3.8 shall be demonstrated at 43.650 MHz.

4.6.7.20 AM hum and noise level. Compliance with the requirements of 3.3.3.9 shall be demonstrated at 43.650 MHz.

4.6.7.21 Tone modulation. Compliance with the requirements of 3.3.3.10 shall be demonstrated at 43.650 MHz.

4.6.7.22 Battery power supply. Compliance with the requirements of 3.3.5 shall be demonstrated. Tests employing batteries for battery life determination will be conducted on first article radio sets only.

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4.6.7.23 Antenna. Compliance with the requirements of 3.3.4 shall be demonstrated.

4.6.7.24 Audio transducer, internal. Compliance with the requirements of 3.3.6.2 shall be demonstrated.

4.6.7.25 SVM interface - mechanical. Verification of the mechanical interface requirements of 3.3.11 shall be made during inspection (see 4.5) using an SVM, TSEC/KYV-2 or -2A provided as Government-loaned equipment (see 3.14).

4.6.7.26 SVM interface - electrical. First article and production verification of the electrical interface requirements of 3.3.12 shall be made in accordance with Appendix A and Drawing No. 977920 to ensure that the radio set SVM connector has been correctly wired.

4.6.7.27 Covers and seals. Compliance with the requirements of 3.3.9 shall be demonstrated.

4.6.7.28 Receive response time. Compliance with the requirements of 3.3.2.9 shall be demonstrated at 43.650 MHz.

4.6.7.29 Transmit response time. Compliance with the requirements of 3.3.3.12 shall be demonstrated at 43.650 MHz.

4.6.7.30 Audio accessory. Compliance with the requirements of 3.3.6.1 shall be demonstrated.

4.6.7.31 Vehicular power. Compliance with the requirements of 3.3.1.7 shall be demonstrated.

4.6.8 Safety. The contractor shall document design actions introduced to facilitate compliance with requirements of 3.7.

4.7 Inspection of packaging. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packaging and packing. Appropriate military levels of protection or commercial packaging shall be specified in the contract or order (see 6.2). Military levels shall be in accordance with MIL-E-17555 and commercial packaging shall be in accordance with ASTM D3951-82.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers stipulating military packaging shall be marked in accordance with MIL-STD-129. Commercial packaging shall be marked in accordance with ASTM D3951-82.

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6. NOTES

6.1 Intended use. This radio set is intended to be a lightweight, reliable, handheld, pocket-size transceiver whose purpose is to facilitate command and control at lower unit/individual levels. Reliable communications between two AN/PRC-68A Radio Sets is provided at ranges up to 300 m (323 yards) with standard antenna and 1.6 km (1 mile) or more with AT-892/PRC-25 antenna (with flexible base removed) in any amphibious and field environment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is required (see 3.1).
(c) When reliability program plan is required (see 3.4.1).
(d) When reliability prediction is required (see 3.4.2).
(e) When maintainability program plan is required (see 3.5.1).
(f) When maintainability prediction is required (see 3.5.2).
(g) Serial numbers to be used on the nameplates (see 3.12).
(h) When Government-loaned special tooling will be utilized by the contractor for production (see 3.14).
(i) When first article test plan is required, and number of first articles to be tested (see 4.3).
(j) When separate maintainability demonstration plan for first articles is required (see 4.6.3).
(k) When maintainability demonstration is required and may contain the optional depot maintenance portion (see 4.6.3).
(l) Levels of preservation, packaging and packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements Lists (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data identified by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.	Options
4.3	First Article Inspection Procedure	DI-T-4901	-

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3.4.1	Reliability Program Plan	DI-R-7079	-
3.4.2	Report, Reliability Prediction	UDI-R-22583A	-
3.4.1	Failure Analysis and Corrective Action Report	DI-R-5299C	-
3.5.1	Plan, Maintainability Program	DI-R-2127	-
3.5.2	Report, Maintainability Prediction	UDI-R-23567	-
4.6.1.14	Tempest Test Plan	DI-T-1912A	-
4.6.3	Plan, Maintainability Demonstration	DI-R-2129	-

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commerical testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the procurement contracting officer.

6.4 Government-furnished property. The procurement contracting officer should arrange to furnish the property listed in 3.13.

6.5 Government-loaned property. The procurement contracting officer should arrange to loan the property listed in 3.14.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity
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APPENDIX A

INTERFACE BETWEEN
AN/PRC-68A RADIO SET AND TSEC/KYV-2 SVM

10. SCOPE.

10.1 Scope. This document specifies the electrical and mechanical interface requirements for the AN/PRC-68A Radio Set when used with the TSEC/KYV-2 or -2A SVM.

20. APPLICABLE DOCUMENTS

20.1 Government Documents.

20.1.1 Drawings. NSA

ON220969	Screw, Externally Relieved Body
ON221004	Connector and Interface Connection Details
ON221019	Connector, Filter RFI
ON221023	Case, Secure Voice Module
ON221024	Plate Mounting, Circuit Card and Connector

30. INTERFACE

30.1 General. The TSEC/KYV-2 or -2A SVM is a self-contained speech encryption device designed for mechanical attachment to the AN/PRC-68A Radio Set. The SVM accepts plain text analog voice from the radio, digitizes this information, encrypts the digital data and returns the digital data (at a 16 kilobits per second (kb/s) rate) to the radio set for transmission. In the receive mode, the SVM accepts cipher text information from the radio, reconstructs the data and clock from this information, decrypts the data, converts the decrypted data to analog information and supplies this analog information back to the radio for amplification as plain text analog speech. The nominal SVM dimensions are 8.89 cm (3.50 inches) high, 9.53 cm (3.75 inches) wide and 3.81 cm (1.50 inches) deep.

30.2 Electrical interface.

30.2.1 Secure voice interface. The input and output signals, required drive levels, impedances, and pin assignments of the secure voice interface shall be as defined in Drawing No. ON221004.

30.2.2 Plain mode jumper plug. Normal plain mode operation of the radio set shall be achieved by use of the jumper plug, as defined in Drawing No. ON221004. This shall be achieved by inserting the jumper plug in the secure voice interface connector on the radio set.

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30.2.3 EOM. The SVM, at the end of the secure transmission, sends an EOM indication. The time duration of the EOM is approximately 40 ms. The radio set shall be capable of being both in transmit mode and receive mode simultaneously (PTT B+ and Receive B+ lines powered at the same time) during the EOM period. This shall be accomplished by the SVM supplying the PTT B+ power to the radio set during the EOM time period.

30.2.4 Chopped B+ cycle. When the radio set is operating in the "power-saver" mode, i.e. squelch on, and no signal is being received, the power-up or "ON" time of the receiver circuits must be 45 ms or greater. This is to allow adequate time for the SVM to power up and detect an incoming digital signal. To ensure that the receiving SVM does not miss the initial synchronization information sent out by a transmitting SVM, the power-down or "OFF" time of the Chopped B+ cycle shall not be greater than 175 ms.

30.2.5 Audio isolation. To prevent distortion of the received digital signal at the output of the radio detector, there shall be adequate isolation between the radio set detector circuits and the radio set audio circuits. This isolation shall be in excess of 30 dB as measured at pin 5 with a 300-Hz audio signal applied to pin 4 of the radio set secure voice interface connector, and radio volume at full level and the radio quieted with a suitable RF carrier.

30.3 Mechanical interface.

30.3.1 SVM attachment. The radio set bottom surface shall allow attachment of the SVM by means of two long threaded bolts which extend through the case and screw into threaded bosses on the radio set. Refer to Drawing No. ON221023 for dimensions and SVM case details. Refer to Drawing No. ON220969 for dimensions of the threaded bolts. The footprint of the radio set bottom shall match that of the SVM. The location of the secure voice interface connector shall be such that it will mate with the connector on the SVM under worst case conditions (see Drawing Nos. ON221019 and ON221024).

30.3.2 Jumper plug storage. When an SVM is attached to the radio set, storage of the plain mode jumper plug shall be on the radio set surface and shall not interfere with the mechanical mating of the radio set and the SVM.

40. TESTING

40.1 Test fixture. The SVM interface wideband test fixture verifies that all 15 interconnection lines are properly connected within the radio set. The fixture also provides interface jacks for testing the XMT CIPHER, XMT AUDIO, RCV CIPHER, and RCV AUDIO

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lines independently with external test equipment to verify compliance with the electrical interface requirements of the SVM. Verification of the RCV B+, XMT B+, CHOPPED B+, BATTERY B+, and ON/OFF B+ lines are accomplished with LED indicator circuits. The ground return currents for these indicator circuits are used to verify the proper connection of Pins 2, 10, 13, and 8 ground -return wires. The proper connections for the SQUELCH DISABLE and 150 Hz DISABLE lines are verified by proper operation of the fixture disable switches during normal operation of the radio set. Design of the wideband test fixture is given in Drawing No. 977920.

40.1.1 Indicator and switch functions.

40.1.1.1 RCV B+ indicator. The RCV B+ indicator should be ON when the radio set is in the RECEIVE mode and OFF when the radio set is either in the TRANSMIT mode or the OFF position. This indicator verifies the proper connection of Pin 6 (RCV B+) and Pin 2 (GND).

40.1.1.2 CHOPPED B+ indicator. The CHOPPED B+ indicator should be ON continuously when the radio set is in the unsquelched RECEIVE mode. The indicator should be flashing ON and OFF when the radio set is in the squelched RECEIVE mode. The indicator should be OFF when the radio set is in either the TRANSMIT mode or OFF position. This indicator verifies the proper connection of Pin 3 (CHOPPED B+) and Pin 8 (GND).

40.1.1.3 BATTERY B+ indicator. The BATTERY B+ indicator should be ON when the external power supply is ON. The indicator should stay ON when the radio set is ON or OFF or when the radio set is in either the TRANSMIT or RECEIVE modes. This indicator verifies the proper connection of Pin 15 (BATTERY B+) and Pin 10 (GND).

40.1.1.4 XMT B+ indicator. The XMT (Transmit) B+ indicator should be ON when the radio set is in the TRANSMIT mode and remain ON during the EOM (end of message) delayed TRANSMIT mode. The XMT B+ indicator should be OFF when the radio set is in either the RECEIVE mode or OFF position. During the EOM TRANSMIT mode both the XMT B+ and RCV B+ indicators should be ON. This indicator verifies the connection of Pin 12 (XMT B+).

40.1.1.5 ON/OFF B+ indicator. The ON/OFF B+ indicator should be ON when the radio set is in the ON position. The indicator should be OFF when the radio set is in OFF position. The indicator should remain ON when the radio set is in either TRANSMIT or RECEIVE mode. This indicator verifies the proper connection of Pin 11 (ON/OFF B+) and Pin 13 (GND).

40.1.1.6 SQUELCH ON/OFF switch. When the SQUELCH ON/OFF switch is in the ON position, normal receiver squelch operation should be enabled. When the switch is in the OFF position, the normal

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receiver squelch operation should be disabled. Normal operation of this switch verifies the proper connection of Pin 7 (SQUELCH DISABLE).

40.1.1.7 150 Hz ON/OFF switch. When the 150 Hz ON/OFF switch is in the ON position, normal 150 Hz squelch tone modulation of the transmitter carrier should be enabled. When the switch is in the OFF position, the normal 150 Hz tone should be disabled. Normal operation of this switch verifies the proper connection of Pin 9 (150 Hz DISABLE).

40.1.1.8 RCV WB/NB switch. When the RCV WB/NB switch is in the NB (narrowband) position, the RCV CIPHER and RCV AUDIO lines should be jumpered for normal RECEIVE mode operation. When the switch is in the WB (wideband) position, the lines should be isolated to verify the absence of an unintentional short circuit between lines within the radio set. The WB position of the switch also connects the RCV CIPHER line and RCV AUDIO line to the WB Audio Output and NB Audio Input jacks respectively for special SVM interface testing. Normal operation of this switch function and the special interface tests verify the proper connection of Pin 5 (RCV CIPHER) and Pin 4 (RCV AUDIO), and the proper operation of the receiver IF-AF circuitry for wideband operation with the SVM.

40.1.1.9 XMT WB/NB switch. When the XMT WB/NB switch is in the NB position, the modulator XMT CIPHER and XMT AUDIO lines are jumpered for normal radio set operation. When the switch is in the WB position, the lines are isolated and routed to the WB Output and NB Input test jacks on the fixture. These jacks provide the interface for testing the modulator speech amplifier (NB Output) and deviation oscillator (WB Input). Normal operation of this switch function in conjunction with the external test jack interface verify the proper connection of Pin 1 (XMT CIPHER) and Pin 14 (XMT AUDIO) and the proper operation of the transmitter modulator circuitry for wide-band operation with the SVM.

40.1.1.10 EOM-HOLD switch. When the EOM switch is in the up position, normal radio set transmit-receive operation is enabled. When the EOM switch is held in the down position, the radio set is in the transmit-EOM mode during which time both the RCV B+ and XMT B+ lines are energized and the CHOPPED B+ line is deengerized.

40.1.1.11 CHOPPED B+. The CHOPPED B+ signal is a periodic pulse with a nominal ON (B+ V_{dc}) time of 50 ms and a nominal OFF time of 150 ms. The ON time shall not be less than 45 ms. The OFF time shall not be greater than 175 ms.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-29401B(MC)	RADIO SET: AN/PRC-68A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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